NOT MEASUREMENT SENSITIVE

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30 September 2008
SUPERSEDING
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PERFORMANCE SPECIFICATION

GREASE, AUTOMOTIVE AND ARTILLERY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of a multi-purpose grease for lubrication of ground vehicles and equipment (see 6.1) and is identified by Military Symbol GAA and NATO Code Number G-403.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSRD-TAR-E/268, MS-268, Warren, MI 48397-5000 by letter or emailed to dami_standardization@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil

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2.2 Government documents.

2.2.1 Government documents, drawings, and publications. The following Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

STANDARDS

FEDERAL

FED-STD-791 - Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Service, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094, or at http://assist.daps.dla.mil/quicksearch).

U.S. DEPARTMENT OF LABOR (DOL)

OSHA 29 CFR 1910.1200 - Hazard Communication; Interpretation Regarding Lubricating Oils.

(Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington DC 20402, or at www.dol.gov).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD Adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D217 - Cone Penetration of Lubricating Grease.
D1092 - Apparent Viscosity of Lubricating Greases
D1831 - Roll Stability of Lubricating Grease.
D2265 - Dropping Point of Lubricating Grease Over Wide Temperature Range.
D2266 - Wear Preventive Characteristics of Lubricating Grease (Four-Ball Method).
D2596 - Measurement of Extreme Pressure Properties of Lubricating Grease (Four-Ball Method).
D3527 - Life Performance of Automotive Wheel Bearing Grease.
D4048 - Detection of Copper Corrosion from Lubricating Grease by the

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Copper Strip Tarnish Test.
D4057 - Manual Sampling of Petroleum and Petroleum Products.
D4289 - Testing Compatibility of Lubricating Greases with Elastomers.
D5483 - Oxidation Induction Time of Lubricating Greases by Pressure Differential Scanning Calorimetry (PDSC).
D5864 - Determining Aerobic Aquatic Biodegradation of Lubricants or Their Components.
D5969 - Corrosion-Preventive Properties of Lubricating Greases in Presence of Dilute Synthetic Sea Water Environments
D6184 - Oil Speration from Lubricating Grease (Conicial Sieve Method).
D6731 - Determining the Aerobic, Aquatic Biodegradability of Lubricants or Lubricant Components in a Closed Respirometer.
D7373 - Predicting Biodegradability of Lubricants using a Bio-kinetic Model.
D7342 - Shear Stability of Lubricating Grease in Presence of Water (Water Stability Test).
E145 - Gravity-Convection and Forced-Ventilation Ovens.
E1131 - Composition Analysis by Thermogravimetry (TGA).

(Application for copies may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959, or at www.astm.org).

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Greases furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.1.1 and 6.4).

3.2 Materials. Grease should be made with biodegradable material and a lithium complex thickening system capable of meeting the requirements herein. If a candidate grease has a different thickening system, it must be compatible with lithium complex thickened grease.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Operating requirements. The grease shall meet all the operating requirements as specified in 3.3.1.

3.3.1 Finished grease properties. The values, after the application of tolerances

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(see 4.1.1.1), shall not exceed the maximum (max.) or fall below the minimum (min.) limits as specified in Table I.

TABLE I. Finished grease properties.

Characteristics	Limits
Dropping point, °C, minimum 1/	220
Worked penetration (1/10 mm)	265-295
Oxidation stability (PDSC) at 210 ºC, minimum, min	15
Corrosiveness (copper strip), maximum 2/	1 b
Water stability, after 10 0000 Double strokes, +10.0% water, Penetration (1/10 mm),	-25 to +60
Life performance (four test runs) at 160°C, hours, minimum	100
Evaporation loss (TGA) at 180°C, 1 hr - Percent, maximum	10.0
Oil separation @ 180 ºC, %, maximum	8.0
Wear preventive characteristics - Avg. scar dia. mm, maximum	0.60
Load carrying capacity – Load wear index, kgf, minimum Welding point, kg, min	 35.0 250
Low temperature torque, N·m, maximum at -54°C Breakaway Running (at 5 min)	 7.00 5.00
Worked stability, after 10 0000 Double strokes, penetration (1/10 mm),	 -25 to +60
Elastomer compatibility CR: Volume change, % Hardness change, Durometer-A points NBR-L: Volume change, % Hardness change, Durometer-A points	 0 to + 40 0 to – 15 -5 to +30 +2 to -15
Salt water corrosion resistance	Pass
Roll stability, penetration (1/10 mm), Minimum/maximum	-25 to +60
Biodegradability, minimum, %	60

1/ °C = degrees Celsius

2/ The grease shall show no green color in that portion contacting the copper strip. The copper strip shall not tarnish more than a classification of l b when compared with the ASTM copper strip corrosion standards (ASTM D4048).

3.4 Support and ownership requirements. The grease shall meet all support and ownership requirements as specified in 3.4.1 through 3.4.5.

3.4.1 Hazardous materials. The grease shall not contain carcinogenic or potentially carcinogenic constituents as defined under the Hazard Communication Standard 29 CFR 1910.1200.

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3.4.2 Storage stability. The grease shall be capable of prolonged storage without degradation in performance and any sign of biodegradation.

3.4.3 Color. The grease color shall be the discretion of the manufacturer as long as the grease contains no dye which could cause discoloration to adjacent surfaces, personnel or clothing.

3.4.4 Compatibility. The grease shall be compatible with all qualified greases when tested in accordance with 4.3.8.

3.4.5 Odor. The grease shall have no odor, which by its rancid, alcohol, or perfume nature, will impede its application or use by personnel.

3.4.6 Homogeneity. The grease shall be homogeneous, smooth in texture, and free of entrapped air which could cause variation in its performance.

4. VERIFICATION

4.1 Classification of inspection. Inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.1.1).
b. Conformance inspection (see 4.1.2).

4.1.1 Qualification inspection. Qualification inspection shall consist of all the verifications listed in table II and shall be in accordance with acceptable tolerances (see 4.1.1.1).

4.1.1.1 Tolerances. Acceptable tolerances for values or limits in 3.3.1 shall be as specified by the qualifying activity (see 6.2).

TABLE II. Verification methods

Title	Requirements	Verification
Operating requirements	3.3	4.3.3
Finished grease properties	3.3.1	4.3.3.1
Compatibility	3.4.4	4.3.8
Support and ownership	3.4	4.3.9
Hazardous materials	3.4.1	4.3.9.1
Storage stability	3.4.2	4.3.9.2
Color	3.4.3	4.3.9.3
Odor	3.4.5	4.3.9.4
Homogeneity	3.4.6	4.3.9.5

4.1.2 Conformance inspection. Unless otherwise specified (see 6.2 and 6.4), the results obtained shall fall within the tolerances/specific values determined at the time of qualification (see 4.1.1.1). The grease shall be tested for all the requirements specified in table II, except for elastomer compatibility, life performance, low temperature torque, roll stability from table I and

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storage stability.

4.2 Sampling for tests. The sample for tests shall consist of two 2.27 kilogram (kg) samples of grease taken at random from filled containers from each lot of grease (see 6.3). For users who obtain grease in large containers, two 2.27 kg samples shall be taken in accordance with ASTM D4057. The lot shall be unacceptable if either sample fails to comply with any of the tests specified.

4.3 Verification methods. The types of verification methods included in this section are visual inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously approved or previous qualified designs.

4.3.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost effective sampling procedures, to verify performance. See the contract for alternatives that replace verification required by this specification.

4.3.2 Inspection conditions. Unless otherwise specified herein, tests shall be conducted on unworked grease, in any sequence.

4.3.3 Operating requirements verification. Complete each test under 4.3.3.

4.3.3.1 Finished grease property verifications. Use the test methods as specified in table III to determine that finished grease properties are as specified within tolerances (see 4.1.1.1).

Table III. Finished grease property test methods.

Test	ASTM and FTM Method
Dropping point	D2265
Oxidation stability (PDSC) at 210°C	D5483
Worked penetration	D217
Corrosiveness (copper strip)	D4048
Water stability	D7342 Procedure A
Life performance	D3527
Evaporation (TGA)	E1131
Wear preventative characteristic	D2266
Load carrying capacity	D2596
Oil separation	D6184
Low temperature torque	FTM 7501 1/
Worked stability	D217 2/
Salt water corrosion resistance	D5969 3/
Elastomer compatibility	D4289
Roll stability	D1831
Biodegradation	D5864, D 6731 or D7373 4/

1/ Low temperature torque shall be performed in accordance with 4.3.4

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2/ Worked stability shall be performed in accordance with 4.3.5.
3/ Salt water corrosion resistance test shall be performed in accordance with 4.3.6.
4/ Biodegradation test shall be performed in accordance with 4.3.7.

4.3.4 Low temperature torque. The test shall be conducted at -54 ±0.5°C according to the Federal Test Method 791, 7501. The breakaway (maximum) torque shall be determined at the beginning of the operation and the running torque shall be determined after five minutes of operation. Both determinations should be calculated by the averaging of four test runs. If the test apparatus is not available, the ASTM D1092 test method shall be used to determine the breakaway torque at -54°C as follows: The apparent viscosity shall be determined at shear rate, 25 reciprocal seconds. This value is then converted to the breakaway torque Newton-meters (N·m) using the following correlation equation (1). If this alternate method is used, only breakaway torque values can be reported. However, in case of referee disputes, only the results from the low temperature torque test method shall be accepted. Failure of the grease to meet the requirements of table I shall constitute failure of this test.

$$\text{Breakaway torque (N·m)} = 2.36 \times (\text{apparent viscosity at } 25\ \text{sec}^{-1} \times 10^{-4}) + 1.48 \quad (1)$$

[Standard error: 0.373]

4.3.5 Worked stability test. The test shall be performed using the standard procedure of ASTM D217 with the following modifications. A change in worked penetration of the grease of more than 60 units higher or 25 units lower than the original worked penetration shall constitute failure of this test.

a. Work a sample of the test grease on a motorized grease worker for 10 0000 double strokes (approximately 28 hours) in accordance with the procedure described in ASTM D217 for prolonged worked penetration.\
b. Immediately after the grease has been subjected to 10 0000 double strokes, bring the grease worker and the sample to 25 ±0.5°C in accordance with the procedure described in ASTM D217 for prolonged worked penetration.
c. Immediately after the test grease sample reaches 25 ±0.5°C, determine the prolonged worked penetration of the grease.

4.3.6 Salt water corrosion test. This test shall be performed using the standard procedure of ASTM D5969 with NaCl solution (1 percent NaCl by weight in distilled water). Failure of the grease to meet the requirements as defined in ASTM D5969 under rating shall constitute failure of this test.

4.3.7 Biodegradability. Determine the percent of theoretical CO_2 evolved (P) of the grease IAW ASTM D 5864 or O_2 consumption (D) according to ASTM D 6731 or the percentage of biodegradability can be determined according to ASTM D7373. A value of biodegradability less than 60 % in 28 days shall constitute failure of this test.

4.3.8 Compatibility. Two grease samples shall be mixed using an electric mixer in 10 %, 50 %, and 90 % ratios. These mixture samples must be stored at room temperature for seven days. Then, the mixture samples shall be tested according to Table III except for biodegradation test. Failure of each mixture sample to meet the requirements of table I shall constitute failure of

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this test.

4.3.9 Support and ownership requirements. Complete each test under 4.3.9.

4.3.9.1 Hazardous materials and compatibility tests. Use one or more of the methods outlined in 4.3 and 4.3.1 to verify the absence of hazardous materials or hazards to personnel.

4.3.9.2 Storage stability test. The test shall be performed on a 6.8 kg unopened can of grease that has been dated and labeled "Storage Stability Test. The undisturbed can or pail of the test grease shall be placed in a thermostatically controlled, dark, convection oven, meeting the requirements of ASTM E145, located in an area essentially free from vibration, at 38 ±3°C for 180 ±5 days. The can shall be positioned in center of the oven and raised at least 75 mm from the bottom, to ensure temperature uniformity. At the end of the storage interval, the can of test grease shall be subjected to the tests of table III except for biodegradation, life performance, water stability, and low temperature torque. Failure of the grease to meet the requirements of table I, within tolerances, shall constitute failure of this test (see 4.1.1.1).

4.3.9.3 Color tests. Use one or more of the methods outlined in 4.3 and 4.3.1 to verify the absence of any dye which could cause discoloration to adjacent surfaces, personnel or clothing.

4.3.9.4 Odor tests. Use one or more of the methods outlined in 4.3 and 4.3.1 to verify the absence of rancid, alcohol, or perfume odor, which could impede application or use of the grease.

4.3.9.5 Homogeneity tests. Use one or more of the methods outlined in 4.3 and 4.3.1 to verify the absence of entrapped air and the homogeneous and smooth texture of the grease.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The grease covered by this specification is intended for the lubrication and surface corrosion protection of all ground vehicles and equipment operated over the temperature range from -54 to +180°C. This grease may also be used in other applications

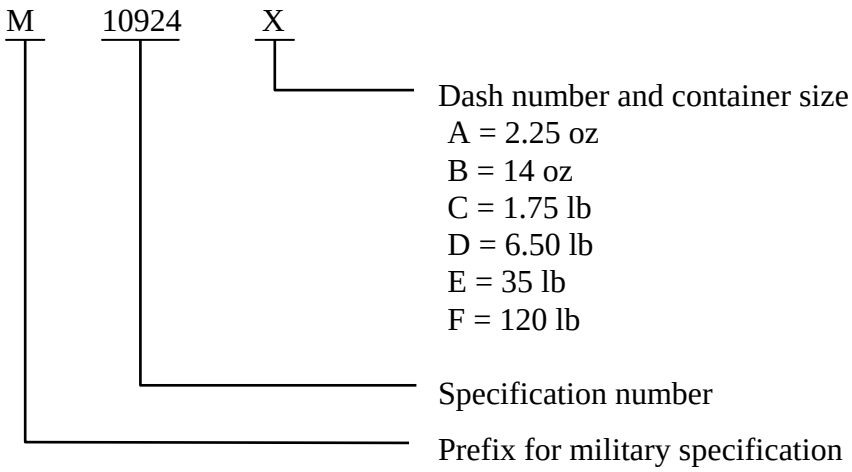
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within this temperature range where a National Lubricating Grease Institute (NLGI) No. 2 consistency grease with oxidation resistant and corrosion prevention properties is desirable. This grease is not intended for use on machinery which comes in contact with food. This specification requires a grease that is capable of lubrication and corrosion protection over an extremely wide range of temperatures; there is no currently available non-government standard which fulfills the requirements of this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. The specific issue of individual documents referenced (see 2.3).
c. Quantity of grease requested, in kilograms.
d. Size and type of container for grease.
e. Tolerances for finished oil requirements (see table I and 4.1.1.1).
f. If conformance testing is other than as specified (see 4.1.2).
g. Packaging requirements, quantity and PIN (see 5.1 and 6.3).

6.3 Part or identifying number (PIN) configuration. The PIN to be used for oils acquired to this specification are created as follows:



6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 10924 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from US Army Tank-automotive and Armaments Command, Warren, Michigan 48397-5000.

6.5 Conformance testing. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, government schedules and budget, product material and design

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maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table II and their designated frequency based on a risk assessment for the procurement.

6.6 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313; and 29 CFR 1910.1200 requires that the Material Safety Data Sheet for each hazardous chemical used in an operation must be readily available to personnel using the material. Contracting officers will identify the activities requiring copies of the Material Safety Data Sheet.

6.7 National stock numbers. The following National Stock numbers have been assigned to the grease covered by this specification:

Grease	National Stock No.
Tube, 2-1/4 oz. (64 g)	9150-01-197-7688
Cartridge, 14 oz. (400 g)	9150-01-197-7693
Can, 1.75 lb (800 g)	9150-01-197-7690
Can, 6.50 lb (3 kg)	9150-01-197-7689
Pail, 35 lb (16 kg)	9150-01-197-7692
Drum, 120 lb (54 kg)	9150-01-197-7691

6.8 Subject term (key word) listing.

Grease
Corrosion, surface protection
Lubricating grease, ground vehicles
(NLGI), No. 2 consistency
Biodegradability

6.9 International Standardization Agreement. Certain provisions of this specification are subject to international standardization agreement STANAGs 1135 and 2845. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Army - AT

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Review Activities:
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